IMPLEMENTATION AUDIT / OCTOBER 2026

The Dragon Passion Problem at Pulp Riot Juice Co.

Full, code-verified review of the synthetic data, assumptions, hypotheses, decision rules, loss attribution, replay and sensitivity.

50 / 50	16 / 16	244	73
reconciliation checks	tests passed	assumption rows	sensitivity metrics

> Blunt overall finding: this is a deterministic, case-fitted prototype, not an empirically estimated or physically integrated supply-chain simulation. Its reconciliation is strong, but many causal allocations and counterfactual benefits are assumptions by construction.

Prepared from the actual repository code, configuration and generated CSVs. Regenerated datasets matched the included CSVs byte-for-byte. The requested 244-row assumptions register is supplied separately as assumptions_register.csv.

Audit basis

I inspected the generator, config, generated CSVs, analytical functions, UI controls, reconciliation code and tests. I also regenerated the data, reran reconciliation and tests, and added reproducible sensitivity runs.

- Regenerated CSVs were byte-for-byte identical to the included files.
- Reconciliation result: 50 PASS / 0 FAIL.
- Automated tests: 16 passed.
- Sensitivity output: 73 result rows in sensitivity_results.csv.
- Assumptions register: 244 unique rows in assumptions_register.csv, with no blank required fields.
- README.md links the register and sensitivity artifacts.

> Source caveat: the original case document is not in the repository. CASE-STATED means ‘marked as a case input in this repository’, usually in config/case.json or a config row with source=CASE. The classification could not be independently checked against the original McKinsey document.

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1) Synthetic data: how it was made

1.1 Common demand-generation method

The generator uses one NumPy random generator with seed 42, covering 1 January–30 June 2026. The year and seed are assumptions, not case facts. Code: src/data_generator.py::SyntheticDataGenerator.__init__; config/assumptions.csv A001–A002.

For each flavor × channel, daily latent-demand weights are:

```
weight = monthly seasonality
       × weekend factor
       × lognormal noise
       × viral multiplier
```

- Monthly multipliers: Jan 0.95, Feb 1.05, Mar 0.90, Apr 1.00, May 1.10, Jun 1.00.
- Saturday/Sunday uplift: +6%; there is no other day-of-week pattern.
- Daily noise: numpy.random.lognormal(mean=0, sigma=0.08). A004 calls this coefficient of variation, but the code uses it as log-space standard deviation.
- There is no trend term, region-specific seasonality or channel-specific calendar pattern.
- DP pre-launch weights are zero.
- Integer bottles are assigned using capped largest-remainder apportionment, preserving requested totals exactly.

Code: src/data_generator.py::SyntheticDataGenerator._daily_weights and _build_demand_and_sales; src/utils.py::apportion; config/monthly_seasonality.csv. Apportionment means noise changes the shape of demand, not the prescribed flavor × channel totals.

1.2 Dataset-by-dataset construction

Dataset	Date range and granularity	Exact construction and case anchors	Code
Sales — daily_sales.csv	24,435 rows = 181 days × 5 flavors × 3 channels × 9 depots. Daily, 1 Jan–30 Jun 2026.	Jan–Feb latent demand fixed at 26 m; Mar–Jun at 96 m. Summer flavor shares 38/20/16/14/12%; channel shares 55/25/20%, applied independently to every flavor. Existing-flavor preseason shares are renormalized over 88%. Daily date × depot allocations are jointly apportioned.	SyntheticDataGenerator._build_demand_and_sales; config/flavors.csv, channels.csv, depots.csv; A003
Losses and realised sales — daily_sales.csv	Same grain and dates.	Event-risk-weighted latent demand is scaled to exactly 7.68 m = 8% of 96 m. sales = true_demand − lost_sales; stockout_flag = lost_sales > 0. Losses are not calculated from production, finished-goods inventory or depot stock.	SyntheticDataGenerator._allocate_case_losses and _build_demand_and_sales
GT distributor orders — daily_sales.csv	Daily GT rows.	round(latent_GT_demand × (1 + Normal(0, 0.15))), floored at zero and realised sales. Proxy generated from synthetic ground truth, not independent distributor data.	SyntheticDataGenerator._build_demand_and_sales; A025
QC feed — qc_sell_through.csv	8,145 rows = 181 days × 5 flavors × 9 depots. Daily QC, 1 Jan–30 Jun.	Exact copy of QC realised sales and stock-out flags; no extra noise or missingness. Marketing date = event date; planner consolidated availability = month-end.	SyntheticDataGenerator._build_qc_feed
Monthly forecast — monthly_forecast.csv	270 rows = 6 months × 5 flavors × 9 depots. Monthly, Jan–Jun.	Current actual monthly sales × alternating ±30%, forcing WAPE to 30%. Despite MONTHLY_LAST_YEAR_MIX, there is no last-year dataset or forecast model.	SyntheticDataGenerator._build_monthly_forecast; A062–A063
Depot stock — depot_stock_weekly.csv	1,215 rows = 27 snapshots × 5 flavors × 9 depots. 1 Jan–29 Jun: 1 Jan plus Mondays. Available one day later.	Stock = round(forward 21-day true demand × event/depot factor × lognormal(0, 0.10)); not actual last-year mix. Channels aggregated. Near 30 Jun, missing future days are padded with earlier dates.	SyntheticDataGenerator._build_depot_stock; A019–A024
Write-offs — depot_writeoffs.csv	8 rows, all 30 Jun: MPB and GC in four OVERSTOCK-profile depots.	Fixed ₹6.00 cr allocated using depot demand_weight × flavor summer_share, rounded with final-row adjustment. Not calculated from expiry, stock age or shelf life.	SyntheticDataGenerator._build_writeoffs; A068
Production — production_daily.csv	905 rows = 181 days × 5 flavors. Daily, 1 Jan–30 Jun.	Jan–Feb production equals latent demand by flavor. Each summer month targets 25 m × 95% = 23.75 m, giving 95 m. April DP weight ×1.20; May–Jun PP ×0.50 and DP ×1.50, then normalized.	SyntheticDataGenerator._build_scheduled_production; A015–A018, A057, A069
Capacity — plant_copacker_capacity.csv	18 rows = 6 months × 3 assets. Monthly, Jan–Jun.	Two plants each get 12.5 m monthly capacity and 50% of actual output. Summer capacity 100 m, production 95 m, utilisation 95%. Co-packer 2.5 m/month, 14-day notice, 25% premium, but generated actual output is zero.	SyntheticDataGenerator._build_capacity; A058
Pulp inventory — pulp_inventory_daily.csv	1,448 rows = 181 days × 8 pulps. Daily, 1 Jan–30 Jun.	Consumption = actual bottles × BOM kg/bottle. Non-passion pulps replenish daily and never constrain production. Passion starts at 21 days of early-summer baseline and declines to 14 at closure. Closing = opening + inbound − consumption.	SyntheticDataGenerator._constrain_and_build_pulp; config/bom.csv and pulps.csv; A014, A026
Pulp inbound — pulp_inbound.csv	1,425 shipment rows. Arrivals 1 Jan–30 Jun; bookings from 13 Nov 2025.	Regular receipts smoothed into daily arrivals. Booking = arrival − lead time. Cost = quantity × assumed normal cost × mode multiplier. No supplier capacity.	SyntheticDataGenerator._constrain_and_build_pulp; A056
Supplier/routes — supplier_routes.csv	8 static rows, one primary supplier per pulp; no dates.	Join of suppliers and routes. Passion: one source, one road, 3-day transit, 60% closure probability. ₹180/kg is synthetic even though the supplier row is marked CASE. Imported dragonfruit uses 49-day midpoint assumption.	SyntheticDataGenerator.generate; config/suppliers.csv and routes.csv
Viral/events — viral_profile.csv and event_calendar.csv	Viral: 15 daily rows, 17–31 Mar. Events: 8 rows, 3 Mar–30 May.	Explicit story-calendar tables generated from the same config and assumptions used elsewhere.	SyntheticDataGenerator._build_viral_profile and _build_events

1.3 Exact story-event injection

Event	Implementation	Source status and code
DP launch	DP demand is zero before 3 Mar 2026, then becomes available.	Date case-marked in case.json and flavors.csv; _build_demand_and_sales
Viral ramp	Starts 17 Mar. DP QC starts at assumed 2.5×, rises linearly to 4.0× on 31 Mar, then remains 4×.	Endpoint/duration case-marked; 2.5 and linear path are MY ASSUMPTION, A042; _daily_weights and _build_viral_profile
Cross-channel propagation	DP GT/MT receive 1 + (QC multiplier − 1) × 0.35; peak non-QC multiplier is 2.05×. Other flavors receive none.	A006, MY ASSUMPTION; _daily_weights
Meaning of 4×	Applies to DP QC daily weights only, not total QC/DP/company demand. Season totals are re-apportioned, so the event redistributes volume over time rather than increasing prescribed summer volume.	_daily_weights and _build_demand_and_sales
Planner notice	31 Mar, the same date the viral profile reaches 4×.	A007, MY ASSUMPTION; _build_events
Highway closure	10–30 May inclusive = 21 days. No passion road arrivals 10 May–2 Jun; receipts resume 3 Jun after transit lag.	Dates/duration case-marked; _constrain_and_build_pulp and _build_events
Emergency air	Booked 17 May, arrives 20 May after assumed 3-day lag. Lot = ₹3.2 cr / (₹180/kg × 5) = 35,555.56 kg. Total ₹3.20 cr; normal ₹0.64 cr; premium ₹2.56 cr.	Cost/multiple case-marked; dates, ₹180/kg and interpretation are assumptions A012–A013/A036; _constrain_and_build_pulp
Depot imbalance	From 31 Mar DP factor 0.55. From 10 May onward DP/PP factor 0.10. May–Jun MPB/GC: 2.20× in OVERSTOCK and 0.45× in SHORT depots.	A021–A024, MY ASSUMPTION; _build_depot_stock
Rival launch	28 Apr, exactly 42 days after viral start.	Case says six weeks; exact date synthetic, A067; _build_events

Synthetic production allocation during the closure

When passionfruit is insufficient, the generator gives pulp to DP first, then PP, and reallocates freed bottle volume 60% to MPB and 40% to GC, keeping total output equal to schedule. This hardcoded story rule is separate from the three human allocation options shown in the app. Code: SyntheticDataGenerator._constrain_and_build_pulp; A018.

1.4 Reconciliation results

I ran python generate_data.py --output /tmp/tf-audit-data. The regenerated CSVs matched the included CSVs byte-for-byte. Code: generate_data.py::main; src/reconciliation.py::reconcile.

Check	Actual	Expected	Difference / result
VOL_01 — annual bottles from ₹1,200 cr / ₹60	200,000,000	200,000,000	0 — PASS
VOL_02 — summer latent demand	96,000,000	96,000,000	0 — PASS
VOL_03 — Jan–Feb demand	26,000,000	26,000,000	0 — PASS; expected value is assumption A003
Flavor volumes	MPB 36.48 m; GC 19.20 m; PP 15.36 m; DB 13.44 m; DP 11.52 m	Same	All exact — PASS

Check	Actual	Expected	Difference / result
Channel volumes	GT 52.80 m; MT 24.00 m; QC 19.20 m	Same	All exact — PASS
CAP_01 — Mar–Jun capacity	100,000,000	100,000,000	0 — PASS
CAP_02 — Mar–Jun production	95,000,000	95,000,000	0 — PASS
CAP_03 — utilisation	95.00%	95.00%	0 — PASS
LOSS_01 — censored bottles	7,680,000	7,680,000	0 — PASS
LOSS_04 — realised summer sales	88,320,000	88,320,000	0 — PASS
LOSS_05 — lost share	8.00%	8.00%	0 — PASS
LOSS_02 — lost revenue	₹46.08 cr	₹46.08 cr	0 — PASS
LOSS_03 — lost contribution	₹14.7456 cr	₹14.7456 cr	0 — PASS
FEED_03 — monthly forecast WAPE	30.00%	30.00%	0 — PASS
PULP_02 — passion cover at closure	14.00 days	14.00 days	0 — PASS
EVENT_06 — closure duration	21 days	21 days	0 — PASS
PULP_03 — road arrivals during closure	0	0	0 — PASS
AIR_01 / AIR_02	₹3.20 cr / 5×	₹3.20 cr / 5×	0 — PASS
WRITE_01 — write-offs/discounts	₹6.00 cr	₹6.00 cr	0 — PASS

Overall: 50 PASS / 0 FAIL. Full output: data/reconciliation_checks.csv.

Checks that do not match

There are no failed checks. PULP_01 needs clarification: minimum closing pulp is 0.0543056 kg while the displayed expected value is 0. This is a non-negativity check, not an equality check; reconcile() deliberately overrides its status to PASS when minimum stock is ≥0.

What reconciliation does not prove

- The 96 m, 7.68 m, ₹3.2 cr and ₹6 cr totals are largely imposed by construction.
- Many expected values come from the same config used to generate the data; these are self-consistency checks, not external validation.
- summer_revenue_share = 48% is not read or reconciled, although 48% × 200 m = 96 m happens to hold numerically.
- There is no production → finished-goods inventory → depot → shipment → sales conservation check.
- Entity-count checks compare generated entities with config entities, not independently with the original case document.

2) Assumptions register

Delivered register

The requested exhaustive table is assumptions_register.csv with columns: ID | Assumption | Value used | Source | Why it was needed | Where in code | Impact if wrong.

Source	Rows
CASE-STATED, as encoded in the repository	62
DERIVED FROM CASE	10
MY ASSUMPTION	172
Total	244

Coverage includes the original 154 scalar/row-level assumptions, 48 case.json inputs, 15 additional case-marked table inputs/constraints, 9 derived values and 18 material implicit modeling choices. Rebuild with python build_assumptions_register.py. Impact ratings are audit judgment, not measured elasticities.

Decision-critical entries

This is the readable high-impact subset. The CSV contains every individual BOM, depot, supplier, route, budget, roadmap, seasonality and option row.

ID	Assumption	Value used	Source	Why needed	Where in code	Impact
A003	Jan–Feb baseline	26 m bottles: (200 m − 96 m) / 8 × 2	MY ASSUMPTION	No preseason volume; even spread across eight non-summer months	_build_demand_and_sales	High
DEPOT_D01–D09	Depot identities, regions and weights	D01–D09 weights 14/8/16/9/15/12/10/11/5%	MY ASSUMPTION	Case gives counts, not demand split	config/depots.csv; demand/stock/write-off builders	High
DEPOT_D01–D09 aggregate	Region split	North 22%, West 25%, South 37%, East 16%	MY ASSUMPTION	Depot allocation and regional transfers	config/depots.csv; _build_demand_and_sales	High
MODEL_DP_QC_SHARE	DP's QC share	20% of DP = 2.304 m bottles; 2.4% of total summer	MY ASSUMPTION	Total QC mix does not specify DP channel mix	_build_demand_and_sales	High
MODEL_VIRAL_APPLICATION	Application of 4×	DP QC daily weights, 2.5× → 4× over 14 days, totals renormalized	MY ASSUMPTION	Response path and normalization unspecified	_daily_weights and _build_demand_and_sales	High
A006	Historical GT/MT viral spillover	35% of uplift above baseline	MY ASSUMPTION	Only QC 4× is given	_daily_weights	High
A034	Forward GT/MT scenario spillover	50% of uplift above baseline	MY ASSUMPTION	Weekly reforecast and scenarios	weekly_rolling_reforecast; build_demand_scenarios	High
BOM_001–BOM_011	Pulp-per-bottle conversion	MPB .105 mango/.035 peach/.010 basil; GC .145 guava/.005 chilli; PP .150 passion; DB .090 domestic/.060 imported dragon; DP .054 domestic/.036 imported dragon/.060 passion kg/bottle	MY ASSUMPTION	No BOM coefficients supplied	config/bom.csv; pulp and allocation functions	High
BOM_009–BOM_011	DP passion/dragon split	0.06 kg passion + 0.09 kg dragon per bottle; dragon 0.054 domestic / 0.036 imported	MY ASSUMPTION	DP pulp exposure; 40/60 passion/dragon by mass; dragon 60/40 domestic/imported	config/bom.csv; _constrain_and_build_pulp	High

ID	Assumption	Value used	Source	Why needed	Where in code	Impact
MODEL_UNIFORM_PRICE_MARGIN	Uniform economics	₹60 and 32% applied to every flavor/channel	MY ASSUMPTION	Case gives averages, not a margin matrix	build_loss_attribution; passionfruit_requirement_profile; shortage_allocation_options	High
DERIVED_COPACKER_CAPACITY	Co-packer 10% base	10% × 25 m = 2.5 m bottles/month	DERIVED FROM CASE	Maximum additional capacity	_build_capacity; constrained_weekly_planner	High
MODEL_COPACKER_FLAVOR_FLEXIBILITY	Co-packer flexibility	Implicitly fully flexible; aggregate bottles, no flavor/BOM/changeover restriction	MY ASSUMPTION	No line-eligibility data	constrained_weekly_planner	High
MODEL_CAPACITY_PULP_SEPARATION	Capacity and pulp feasibility	Calculated separately; co-packer volume can appear without simultaneous pulp balance	MY ASSUMPTION	No joint production optimizer	src/analysis.py::build_analysis	High
CASE_PASSION_CLOSURE_PROBABILITY	Annual closure likelihood	60%	CASE-STATED	Expected-loss weighting	closure_loss_scenarios via build_analysis	High
A031–A033	Closure-length weights	1 week 25%, 2 weeks 35%, 3 weeks 40%, conditional on closure	MY ASSUMPTION	No closure-length distribution	build_analysis → closure_loss_scenarios	High
CASE_BOTTLED_SHELF_LIFE_DAYS / DERIVED_MT_QC_PREBUILD	Shelf-life rule	Bottled 90 days; MT/QC require 60 remaining; derived prebuild cap 30 days	DERIVED FROM CASE	Prevent excessive finished-goods prebuild	constrained_weekly_planner	High
—	GT shelf-life rule	Not modeled	—	No GT rule or expiry-lot implementation	No corresponding function	High
A025 / MODEL_GT_ORDER_GENERATION	GT demand proxy	Orders centered on latent GT demand with 15% Normal noise	MY ASSUMPTION	GT outlet data unavailable	_build_demand_and_sales	High
MODEL_CENSORING_CORRECTION	Stock-out correction	In-stock sales unchanged; GT orders if above sales/baseline; otherwise recursive 14-day corrected mean or sales lower bound	MY ASSUMPTION	Latent demand unobserved in stock-outs	estimate_censored_demand; prepare_qc_signal	High
MODEL_LOSS_ALLOCATION	Synthetic stock-out allocation	Risk-weighted rows scaled to exactly 7.68 m, independently of production/depot inventory	MY ASSUMPTION	Case total known; row-level losses unknown	_allocate_case_losses	High
A014	Passion cover before viral ramp	21 days of early-summer baseline requirement	MY ASSUMPTION	Only closure cover is given	_constrain_and_build_pulp	High
A029 + A030	Buffer carrying and storage	18% + 8% = 26% annually of pulp principal	MY ASSUMPTION	No financing or cold-storage rate	build_analysis → calculate_buffer_option	High
MODEL_BUFFER_OBJECTIVE	Buffer objective	Carry/storage + expected contribution loss; air premium excluded	MY ASSUMPTION	Defines optimiser choice	calculate_buffer_option; optimise_buffer	High
SUPPLIER_SUP_PASSION	Normal passion cost	₹180/kg; row also contains case-marked lead/source fields	MY ASSUMPTION	Air-lot size and buffer economics	config/suppliers.csv; _constrain_and_build_pulp	High
A036	Meaning of 5× air cost	5× total normal delivered pulp cost for emergency lot	MY ASSUMPTION	Infer air kilograms from ₹3.2 cr	Hardcoded in _constrain_and_build_pulp; A036 documentary	High
A021–A024	Depot stock factors	2.20 / 0.45 / 0.55 / 0.10 by event and profile	MY ASSUMPTION	Create simultaneous overstocks and shortages	_build_depot_stock	High
A068	Write-off allocation	depot_weight × flavor_share, only MPB/GC in overstock depots	MY ASSUMPTION	Case total known; causal distribution unknown	_build_writeoffs; documentary value, hardcoded logic	High
A035	Transfer sales recovery	70% of transferred deficit bottles become realised sales	MY ASSUMPTION	Convert transfers into expected contribution	suggest_depot_transfers	High

ID	Assumption	Value used	Source	Why needed	Where in code	Impact
A043–A045	Channel delisting weights	GT 1, MT 2, QC 3	MY ASSUMPTION	Relative trade-off score without outlet data	shortage_allocation_options	Med
ALLOC_PROTECT_DP / PROTECT_PP / BALANCED	DP share of available passion	70% / 30% / 55%	MY ASSUMPTION	Three explicit COO choices	config/allocation_options.csv; shortage_allocation_options	High
ATTRIBUTION_*	Lost-contribution shares	Late 35%, route 35%, launch 15%, stale depot mix 15%	MY ASSUMPTION	No measured causal split	config/loss_attribution.csv; build_loss_attribution	High
ATTRIBUTION_*	Air/write-off assignment	100% air to route; 100% write-offs to stale depot mix	MY ASSUMPTION	Root-cause waterfall	build_loss_attribution	High
ATTRIBUTION_*	LOW/BASE/HIGH capture	Late 25/45/65%; route 30/55/75%; launch 40/65/85%; stale 20/40/60%	MY ASSUMPTION	Convert impact into counterfactual avoided loss	build_loss_attribution	High
SUPPLY_OPTION_*	Intervention efficacy	Second supplier 65% loss reduction; co-packer 5%; air trigger 35% loss and 20% air-cost reduction	MY ASSUMPTION	No intervention performance data	supply_option_economics; build_scenario_replay	High
BUDGET_*	Selected budget lines	Build .35; vendor IT .20; cloud .12; training .12; S&OP .10; Excel .08; second supplier .45; air trigger .15; contingency .20 = ₹1.77 cr	MY ASSUMPTION	No vendor quotations	config/budget.csv; build_budget	High
CASE_BUDGET_CAP	First-year cap	₹2.50 cr, hardcoded	CASE-STATED project constraint	Required package constraint	build_budget; budget_reconciliation	High
A037/A038	Scenario uncertainty	±20% fresh / ±40% degraded	MY ASSUMPTION	Not estimated from live forecasts	build_analysis; feed_fallback_assessment	Med
A039	Minimum active alert history	5 nonzero observations	MY ASSUMPTION	Prevent launch-day tiny-denominator alerts	detect_demand_anomalies	Med
A040/A041/A060	Reforecast/scenario windows	4-week reforecast; 28-day scenario history; 8-week horizon	MY ASSUMPTION	Planning-window choices	weekly_rolling_reforecast; build_demand_scenarios; A060 documentary	Med
MODEL_APPROVAL_CHAIN	Approval and escalation	Planner → S&OP → COO for peak trade-offs	MY ASSUMPTION	Proposed governance design	build_risk_register; apply_decision	Med

Approval caveat: apply_decision() checks only that actor text is nonempty. It does not authenticate the user, verify their role or require sequential Planner → S&OP → COO approvals.

3) Hypotheses the model tests

Hypothesis	How the model tests it and observed result	What would prove it wrong / audit verdict
QC leads GT demand	weekly_rolling_reforecast() uses current-week QC uplift versus a prior four-week baseline to modify next-week GT/MT. The generator applies QC and GT/MT effects on the same dates, with no estimated lag. Weekly WAPE is 10.48% against corrected synthetic demand, not GT outlet truth.	Real GT data showing no lead, negative lag or no improvement over a non-QC baseline. Not proven; lead is assumed, not estimated.
Sensing would catch the spike before planners	detect_demand_anomalies() compares daily QC with shifted rolling baselines. Base: Amber 18 Mar, Red 19 Mar, versus assumed notice 31 Mar. Headline 12 days uses Red in preference to earlier Amber.	No alert before the real notice or different ramp/feed latency. Sensitivity shows Red disappears under less front-loaded ramps. Illustrative, not independently validated.
Frozen-pulp buffer beats air freight	optimise_buffer() searches 0–8 weeks. Base optimum +1 week, only ₹0.0946 cr net expected benefit. Expected air premium is displayed but excluded from the objective.	Carry/working capital exceeding avoided total cost, lower closure probability or cheaper/reliable air. Adding displayed air premium changes optimum to +2 weeks. Full buffer-versus-air comparison is not implemented.
Co-packer fixes demand spike but not pulp shortage	constrained_weekly_planner() adds up to 0.575 m/week after notice. In 8-week VIRAL_4X, unmet falls from 13.773 m to 10.323 m but is not eliminated. Pulp is separate.	Flavor ineligibility/changeovers could worsen capacity results; joint model could alter them. Helps capacity but does not fully fix 4×; 'cannot create pulp' is structurally true but not jointly simulated.
Last-year depot mix caused ₹6 cr write-offs	Stock is labeled last-year mix and all ₹6 cr assigned to stale mix. But stock uses forward true demand × factors, and write-offs are allocated by assumption.	Actual ledger showing other depots/flavors/causes, or current-mix simulation not reducing write-offs. Not tested causally; conclusion built in.
Launch handoff made DP unready	launch_readiness_gate() requires every BOM pulp to cover lead + 7 days using forward scheduled requirements. Imported dragonfruit 49 + 7 = 56 days; result BLOCKED; notice/lead gap 35 days.	Actual lead ≤14 days, sufficient 56-day cover or approved alternate supplier. Depends on 49-day midpoint and synthetic inventory.
Viral demand eroded passion cover	build_causal_links() compares old-baseline and current-demand cover at closure: 2.00 → 1.719 weeks. VIRAL_4X increases weekly passion requirement 31.02%.	Actual pulp demand not increasing, alternate BOM or replenishment maintaining cover. Mechanically demonstrated using assumed BOM/spillovers.
Stock-outs censor demand and correction helps	Exactly 7.68 m losses imposed. censoring_validation() reports 34.93% lower absolute error than raw sales on stocked-out rows.	Real-data back-test where proxy does not beat sales or is biased. Estimated lost demand is 9.965 m versus true 7.680 m, a 29.75% overestimate; helps does not mean recovers truth.
Protecting DP improves contribution/kg with PP/rival trade-off	DP uses 0.06 kg/bottle, PP 0.15. shortage_allocation_options() ranks contribution and channel-weighted risk but chooses no winner.	Different actual margins/BOM or channel/rival evidence. Rival shelf-space loss is not quantified.
Proposed workflow can capture modeled benefits	App demonstrates recommendation, approval and audit-log mechanics. Avoided loss then applies assumed capture rates.	Planner non-adoption, missed SLAs, approval delays or lower measured pilot capture. Software mechanics tested; organisational capture not tested.

4) Model parameters and decision rules

UI tunability

Only six detector inputs are real analytical UI controls: baseline window, Amber multiplier and consecutive days, Red multiplier and consecutive days, and minimum active history. Code: app.py::main.

The Architecture page’s QC/depot checkboxes call only feed_fallback_assessment() for a demonstration; they do not rebuild the analytical context. Code: src/pages/budget_architecture.py::render_architecture. Everything below otherwise requires config/code changes, and several documentary config values are not read at all.

Demand, sensing and launch rules

Rule	Value and reason	UI-tunable?
Detector baseline	Trailing 14-day mean, shifted one day. EWMA and z-score are diagnostics, not triggers. Case-encoded threshold; shift is implementation choice. detect_demand_anomalies()	Yes
Minimum history	5 prior nonzero observations, preventing pre-launch zeros from creating false spikes. A039.	Yes
Amber	Signal/baseline >1.5× for 2 consecutive days. Case-encoded.	Yes
Red	Signal/baseline >2.0× for 3 consecutive days. Case-encoded.	Yes
First actionable lead	Red preferred over Amber rather than earliest. Base lead is 12 days although Amber was 13 days before notice. build_causal_links(); build_scenario_replay()	No
Stock-out correction window	Recursive trailing 14 corrected observations. GT proxy must exceed sales and baseline; otherwise rolling baseline or sales lower bound. estimate_censored_demand()	No
Weekly reforecast	4-week rolling baseline; QC uplift clipped to at least 1; GT/MT receive 50% of uplift above 1. weekly_rolling_reforecast()	No
Forward scenarios	28-day history, 8-week horizon; BASE/2×/4× apply to DP QC, with 50% non-QC spillover. build_demand_scenarios()	No
Scenario uncertainty	±20% fresh, ±40% when QC or depot data is missing. A037–A038.	No
Launch gate	Every BOM pulp: opening kg / forward scheduled daily requirement ≥ supplier lead + 7 days; all must pass for READY. Imported dragonfruit 49 + 7 = 56 days. launch_readiness_gate()	No
Launch notice comparison	14-day notice versus 49-day imported lead gives 35-day gap; informational, not an additional gate by itself.	No

Supply, capacity, buffer and allocation rules

Rule	Value and reason	UI-tunable?
Exposure fragility	closure probability × route factor × source factor; factor 1 for single, 0.5 otherwise. RED if cover < lead or fragility ≥0.50; AMBER if cover < lead + 1 week or fragility ≥0.20. build_exposure_map()	No
Closure scenarios	60% annual probability × conditional 25/35/40% for 1/2/3 weeks. Shortage = requirement − stock − buffer, floored at zero. closure_loss_scenarios()	No
Air scenario cost	Shortage kg × ₹180 × 5 total; premium uses ×4. Next-summer scenario does not reconcile to actual ₹3.2 cr lot.	No
Buffer search	Integer 0–8 weeks, ≤180 days. Principal = buffer kg × ₹180. Annual cost = principal × 26%. A046/A029/A030.	No
Buffer objective	Minimise annual carry/storage + expected contribution loss. Expected air premium excluded. Working capital outside selected budget. calculate_buffer_option(); optimise_buffer()	No
Weekly in-house capacity	25 m × 7 / 30.4375 = 5.749 m bottles/week. Historical run rate 95%, leaving 5% headroom. constrained_weekly_planner()	No
Co-packer	10% × 25 m = 2.5 m/month = 0.575 m/week. Eligible after 14 days. Premium = 25% × ₹40.80 = ₹10.20/bottle on used volume.	No
Shelf-life rule	MT/QC prebuild displayed as 90 − 60 = 30 days; structural buffer directed to 180-day frozen pulp. Returned rule/text, not dynamic lot-expiry constraint.	No
Synthetic historical allocation	DP receives pulp first, PP second; freed volume 60% MPB / 40% GC so total output stays at schedule. _constrain_and_build_pulp()	No
Human shortage options	DP pulp share 70% / 30% / 55% for Protect DP / Protect PP / Balanced. Within each flavor, all channels receive same service fraction. shortage_allocation_options()	No
Allocation scoring	Contribution ₹19.20/bottle; unmet risk weights GT 1, MT 2, QC 3. Contribution rank descending, risk rank ascending. No automatic winner.	No
Supply-option economics	Configured assumed reductions. Buffer uses calculated economics; benefits not added to top-down replay savings. supply_option_economics()	No

Inventory, control-tower, replay and governance rules

Rule	Value and reason	UI-tunable?
Current depot cover	Latest snapshot per flavor/depot; trailing 28-day corrected demand; target 21 days. inventory_cover_snapshot()	No
Inventory status	SHORT <14 days; OVERSTOCK >35 days; otherwise WATCH; zero demand gets NO_CURRENT_DEMAND. A052–A053.	No
Stale depot sheet	as_of − snapshot_available_date > 7 days. Stale rows cannot originate transfers. A054.	No
Transfers	Greedy largest-surplus origin to lowest-cover destination; origin retains 21-day cover. Same region 2 days, cross-region 4 days.	No
Transfer acceptance	Quantity × ₹1.50 cost; recovered contribution = quantity × 70% × ₹19.20. Only positive-net suggestions; all require human approval. suggest_depot_transfers()	No
Control-tower demand	Latest DP alert status at page as-of date. Missing alerts produce unavailable/degraded messaging. control_tower_tiles()	No
Inventory tile severity	RED if both SHORT and OVERSTOCK; AMBER if only one or stock unavailable; otherwise GREEN.	No
Distribution tile severity	AMBER when transfers exist or cover unavailable; otherwise GREEN.	No
Transportation tile severity	RED during 10–30 May; otherwise AMBER, even outside closure.	No
Risk severity	R001 RED if alert exists; R002 RED when launch blocked; R003/R004 fixed RED; R005/R006 fixed AMBER. build_risk_register()	No
Replay actions	Manual route input 3 May, allocation deadline 9 May, model air 10 May, actual air 17 May. Assumptions, not learned triggers.	Timeline date only
Replay air cost	₹3.20 cr × (1 − 20%) = ₹2.56 cr. Not added again to avoided loss.	No
Replay buffer line	Actual cover + selected buffer weeks; static stress-test, not re-simulated inventory.	No
Loss-impact LOW/HIGH cards	Base impact ±25% of allocated contribution component only; air and write-offs fixed. A047.	No
Avoided-loss LOW/BASE/HIGH	Uses cause-level capture rates, not the ±25% impact columns. build_loss_attribution()	No
Budget selection	Sum selected config rows; must be ≤₹2.50 cr. Buffer principal and unselected co-packer excluded. build_budget()	No
Payback	12 × selected cost / expected next-summer benefit; simple, undiscounted. budget_impact()	No
Approval workflow	PROPOSED → APPROVED/REJECTED, or MODIFY → PROPOSED_MODIFIED. MODIFY requires replacement text; actor merely nonempty. Execution remains zero. apply_decision()	Yes, action buttons

5) Loss attribution and replay

5.1 Base calculations

```
Lost bottles       = 96.00 m × 8% = 7.68 m
Lost revenue       = 7.68 m × ₹60 / 1 crore = ₹46.08 cr
Lost contribution  = ₹46.08 cr × 32% = ₹14.7456 cr

Economic impact_i = ₹14.7456 cr × lost_contribution_share_i
                  + ₹3.20 cr     × air_freight_share_i
                  + ₹6.00 cr     × writeoff_share_i
```

Lost revenue is allocated for context using the same 35/35/15/15 shares but is not added again to economic impact. Code: src/links.py::build_loss_attribution.

5.2 Exact root-cause allocation

Root cause	Revenue context	Contribution	Air	Write-offs	Economic impact	Capture L/B/H	Avoided LOW	Avoided BASE	Avoided HIGH
Late demand sensing	₹16.128 cr	₹5.16096 cr	0	0	₹5.16096 cr	25/45/65%	₹1.290240 cr	₹2.322432 cr	₹3.354624 cr
Single source/route	₹16.128 cr	₹5.16096 cr	₹3.20 cr	0	₹8.36096 cr	30/55/75%	₹2.508288 cr	₹4.598528 cr	₹6.270720 cr
Launch handoff	₹6.912 cr	₹2.21184 cr	0	0	₹2.21184 cr	40/65/85%	₹0.884736 cr	₹1.437696 cr	₹1.880064 cr
Stale depot mix	₹6.912 cr	₹2.21184 cr	0	₹6.00 cr	₹8.21184 cr	20/40/60%	₹1.642368 cr	₹3.284736 cr	₹4.927104 cr
Total	₹46.08 cr	₹14.7456 cr	₹3.20 cr	₹6.00 cr	₹23.9456 cr	—	₹6.325632 cr	₹11.643392 cr	₹16.432512 cr

Case inputs versus judgment

Case-marked inputs: 96 m summer bottles, 8% loss share, ₹60/bottle, 32% contribution margin, ₹3.20 cr emergency air and ₹6.00 cr write-offs/discounts.

Judgment/model choices: 35/35/15/15 contribution split; 100% of air to route fragility; 100% of write-offs to stale depot mix; every capture rate; adding full ₹3.20 cr air rather than only the ₹2.56 cr premium; treating contribution, air and write-offs as additive and independent; and the resulting ₹23.9456 cr combined-impact convention.

The attribution shares are not estimated from the generator’s A008–A011 event-level stock-out weights. They are a separate top-down allocation.

5.3 Avoided-loss range in scenario replay

```
Avoided_loss_scenario_i = base_economic_impact_i × capture_rate_scenario_i

LOW  total = ₹6.325632 cr
BASE total = ₹11.643392 cr
HIGH total = ₹16.432512 cr
```

These ranges come from capture-rate choices, not statistical confidence intervals. The separate ±25% contribution sensitivity is used for risk-card LOW/HIGH economic-impact values; it is not used in avoided-loss totals. Code: src/links.py::build_loss_attribution; src/replay.py::build_scenario_replay.

5.4 What replay does and does not simulate

Replay includes Amber/Red dates and notice; conditional co-packer notice on 19 Mar and eligibility on 2 Apr; manual route input 3 May; allocation deadline 9 May; counterfactual air booking 10 May, seven days early; a static buffer-cover line; and human-readable action/owner/approver tables.

Replay does not rerun production, inventory, depot allocation or sales under each recommendation. Avoided loss is entirely the top-down attribution calculation above.

```
Counterfactual air cost = ₹3.20 cr × 80% = ₹2.56 cr
```

That ₹2.56 cr is a 20% lower total scenario cost. It happens to equal the actual lot's separately calculated ₹2.56 cr premium, but those are different concepts. Code: src/replay.py::build_scenario_replay; src/data_generator.py::SyntheticDataGenerator._constrain_and_build_pulp.

6) Limitations and sensitivity

6.1 Five weakest assumptions, ranked by headline influence

This ranking is audit judgment, not a fitted elasticity ranking.

Rank	Weak assumption	Why it is weak / what it changes
1	Root-cause shares and intervention capture rates	No measured causal attribution or adoption evidence. Directly determines ₹6.33/₹11.64/₹16.43 cr avoided loss and 1.82-month payback.
2	Front-loaded viral path and planner-notice date	A042 starts at 2.5× and A007 fixes notice at 31 Mar. Together they manufacture the precise Red-on-19-Mar / 12-day-lead story.
3	Closure/buffer economics	Conditional closure weights, ₹180/kg, BOM coefficients and 26% annual carry are assumptions. +1 week has only ₹0.095 cr net benefit and excludes air premium from its objective.
4	Flavor-independent channel mix, viral spillover and GT proxy	DP QC assumed 20% of DP; GT/MT propagation assumed; GT orders generated from latent truth. Affects alert timing, pulp uplift, reforecast accuracy and censoring correction.
5	Depot stock factors and stale-mix write-off allocation	2.20/0.45/0.55/0.10 factors create the inventory story, while 100% of ₹6 cr is assigned to stale mix without ledger or expiry model.

6.2 Sensitivity 1: viral-ramp shape

Only A042, the initial DP QC multiplier, was varied, using the actual generator and detector. Planner notice remains fixed at assumed 31 Mar. Code: audit_sensitivity.py::sensing_sensitivity; results: sensitivity_results.csv.

Initial multiplier	First Amber	First Red	Headline lead: Red if available else Amber	Amber lead
1.0×	22 Mar	Not triggered	9 days	9 days

Initial multiplier	First Amber	First Red	Headline lead: Red if available else Amber	Amber lead
1.5×	20 Mar	Not triggered	11 days	11 days
2.0×	18 Mar	Not triggered	13 days	13 days
2.5× — base	18 Mar	19 Mar	12 days	13 days
3.0×	18 Mar	19 Mar	12 days	13 days

Does the conclusion hold?

- Some Amber warning precedes assumed planner notice in every tested path.
- The specific Red alert and 12-day headline do not hold at initial multipliers of 1.0–2.0.
- Lead values are not monotonic because the headline prefers Red over earlier Amber.
- All leads remain measured against assumed 31 Mar notice. An earlier real notice could eliminate them.

> Early sensing is plausible in the synthetic design, but the precise 12-day claim is not robust or independently evidenced.

6.3 Sensitivity 2: closure probability

Annual closure probability was varied while every other input remained unchanged. Code: audit_sensitivity.py::closure_sensitivity; src/supply.py::optimise_buffer.

Annual closure probability	Zero-buffer expected loss	Optimal buffer	Net expected benefit of optimum
15%	₹0.286 cr	0 weeks	₹0
30%	₹0.571 cr	0 weeks	₹0
45%	₹0.857 cr	0 weeks	₹0
60% — base	₹1.143 cr	1 week	₹0.0946 cr
90%	₹1.714 cr	1 week	₹0.561 cr

The one-week option’s carry break-even probability is approximately 53.91%.

Does the conclusion hold?

- No. At or below 45% in the tested grid, the optimum is no additional buffer.
- At the case-encoded 60%, the one-week decision has a very small margin.
- Adding the already-displayed expected air premium to the objective changes the optimum from 1 week to 2 weeks, with combined cost ₹1.676 cr.

> The buffer conclusion is sensitive both to probability and to the objective definition.

6.4 Sensitivity 3: lost-contribution attribution split

Only the four contribution shares were changed, leaving air/write-off assignments and capture rates intact. Code: audit_sensitivity.py::attribution_sensitivity; src/links.py::build_loss_attribution; src/budget.py::budget_impact.

Contribution split: late/route/launch/stale	Avoided LOW	Avoided BASE	Avoided HIGH	Base payback
35/35/15/15 — base	₹6.326 cr	₹11.643 cr	₹16.433 cr	1.824 months
Equal 25/25/25/25	₹6.399 cr	₹11.717 cr	₹16.506 cr	1.813 months
Route-heavy 15/55/15/15	₹6.473 cr	₹11.938 cr	₹16.727 cr	1.779 months
Sensing-heavy 55/15/15/15	₹6.178 cr	₹11.348 cr	₹16.138 cr	1.872 months
Extreme 0/0/0/100 to stale mix	₹5.109 cr	₹10.058 cr	₹14.847 cr	2.112 months

Does the conclusion hold?

- Positive modeled benefit survives these split variations.
- The precise ₹11.64 cr benefit and 1.82-month payback do change.
- Robustness is limited: fixed air/write-off totals and positive assumed capture rates remain. If adoption/capture is zero, avoided benefit is zero regardless of split.

> The arithmetic is stable across alternative splits, but the business case remains wholly dependent on unvalidated capture assumptions.

6.5 Other material simplifications and gaps

- 1. No integrated physical flow. Sales losses, production cuts, pulp stock and depot stock are separately constructed. No end-to-end finished-goods or transportation conservation model. Code: SyntheticDataGenerator.generate; src/reconciliation.py::reconcile.
- 2. ‘Last-year mix’ is mostly a label. Monthly forecasts use current actual sales × ±30%; depot stock uses forward true demand × factors. No last-year data exists.
- 3. No real GT outlet validation. The 10.48% weekly WAPE and 34.93% censoring improvement are synthetic and should not be compared directly with monthly 30% WAPE as like-for-like accuracy improvements.
- 4. No shelf-life lot simulation. The 30-day MT/QC prebuild rule is displayed, not enforced through expiry lots or channel receipt/rejection transactions. GT rejection rules are absent.
- 5. No joint pulp/capacity/co-packer optimisation. Co-packer flexibility, supplier capacity, changeovers, line eligibility and BOM feasibility are not simultaneously modeled.

- 6. Pulp constraints affect only part of the story. Passionfruit can constrain DP/PP production, but other pulps never constrain output. The blocked DP launch gate does not physically stop generated DP production.
- 7. No full-year simulation. Only Jan–Jun is generated. The 200 m annual volume is an arithmetic anchor, not a simulated annual total.
- 8. No realistic transport execution. No trucks, lane capacities, shipment consolidation, vehicle costs, alternate physical routes or transfer shelf-life checks.
- 9. No weather/landslide prediction. The 3 May route warning is manual scenario text.
- 10. No measured adoption or causal intervention effect. Supplier qualification, air-trigger, co-packer and planner-workflow benefits are assumed.
- 11. Approval governance is a prototype. No authentication, role permission enforcement, immutable audit store or ERP integration.
- 12. Config is not uniformly live. Fourteen scalar rows are documentary rather than read by Python: A036, A055, A056, A057, A059–A065, A067–A069. Changing their CSV values does not change the hardcoded implementation.
- 13. Several case keys are not read: summer_revenue_share, plant_count, region_count, gt_distributor_count, all three shelf-life keys, dragonfruit_domestic_share, and imported-lead minimum/maximum. Related behavior is hardcoded or duplicated elsewhere.
- 14. Some function calls use defaults rather than forwarding config. Capacity, shelf-life and some exposure calculations are not guaranteed to respond consistently to config edits.
- 15. Source labels are not perfectly consistent. SUP_PASSION is marked CASE while containing synthetic ₹180/kg cost, and channel rows marked CASE coexist with separate FEED_GT/FEED_MT assumptions. The new register separates these where possible and identifies mixed rows.

Things a judge could challenge

Challenge	How you should answer
You engineered the data to tell the story.	Yes, it is a deterministic case-fitted prototype. Totals are reconciled by construction. We separate case inputs, derived values and 172 modeling assumptions, and do not claim empirical causality.
Do 50 passing checks prove the model is right?	No. They prove reproducibility and internal numerical consistency. Several checks are self-referential, and there is no independent case-document validation or end-to-end physical conservation test.
Is the 12-day sensing lead manufactured?	It is conditional on assumed 2.5× initial response and 31 Mar notice. Lower front-loading removes Red in our sensitivity. Present it as an illustrative detector result, not a measured historical lead.
Is ₹11.64 cr a guaranteed saving?	No. It is a top-down counterfactual using judgmental attribution and capture rates. It is not a confidence interval or booked benefit; it requires a measured pilot.
Why is one week the optimal buffer?	Only under 60% closure probability, 26% carry rate and contribution-loss-only objective. Below roughly 54%, zero weeks wins; including expected air premium changes the answer to two weeks.
Does co-packing solve the spike?	Not in the 4× scenario. It reduces capacity-only unmet volume but leaves 10.32 m unmet over eight weeks, and cannot create pulp. Flavor flexibility remains unvalidated.
Did stale depot allocation really cause ₹6 cr?	The prototype does not prove that. It allocates the case total to stale mix by assumption. Actual write-off, expiry and depot ledgers are needed to test causality.
What GT visibility do you actually have?	No outlet-level GT truth. Distributor orders, depot secondary sales and QC are proxies. GT orders here are synthetic and must be replaced and validated with real data.
Is everything config-driven?	Entity expansion is tested, but not every parameter is live. The audit register identifies documentary rows, unused case keys and hardcoded rules so we do not overclaim configurability.
Is the approval system production-ready?	It is a decision-log prototype only. It records human decisions and executes nothing, but still needs authentication, role enforcement, secure persistence and operational integrations.

Audit artifacts

File	Purpose
Pulp_Riot_Dragon_Passion_Audit.pdf	This formatted report
assumptions_register.csv	Exhaustive 244-row assumptions register
sensitivity_results.csv	73 long-form sensitivity metric rows
build_assumptions_register.py	Rebuilds the register from actual config and Python function references
audit_sensitivity.py	Reproduces the viral-ramp, closure, attribution and buffer-objective checks
data/reconciliation_checks.csv	Full 50-check reconciliation output

End of report.